

				Momentum Scalping
2026 Q2	David Trullas Vila	Futures Day Trading	2,222.6%	NASDAQ, Discretionary Momentum Scalping
2026 Q1	Rhuthran Rajendran	Forex	102.3%	Forex Pairs (Unspecified Methodology)
Multiple	Chuck Hughes	Stock / Options	300.0%+	Equities, Keltner Channel Options Trend Following
...				

Strategy Breakdown by Trader

An examination of individual methodologies demonstrates that there is no singular path to extracting exceptional returns. Instead, success is derived from the rigid application of a well-defined edge, whether rooted in fundamental positioning, statistical arbitrage, or classical momentum mechanics.

Larry Williams

Larry Williams remains the most legendary figure in the history of the WCTC, having set an unbroken record in 1987 by compounding a \$10,000 starting balance into over \$11 million, representing an 11,376% return. Williams operates primarily as a swing trader, combining fundamental market positioning data with precise technical timing indicators. His core edge relies heavily on the Commitment of Traders (COT) reports, utilizing this data to track the activities of commercial hedgers versus large speculators. By identifying extreme positioning imbalances, Williams anticipates major macro-level trend reversals orchestrated by institutional "smart money". This is frequently combined with inter-market analysis, such as monitoring the relationship between Treasury bonds and the S&P 500, to filter out false signals.

To time his entries within these macro frameworks, Williams developed the Williams %R momentum oscillator. Operating on an inverted scale from 0 to -100, the indicator identifies short-term oversold conditions (below -80) and overbought conditions (above -20) relative to the highest high of a specific lookback period. The strategy seeks to enter pullbacks that occur contrary to the broader structural trend. Regarding exits, Williams employs a variety of mechanisms, most notably the "bailout exit," which dictates closing a position at the very first profitable daily open, prioritizing a high win rate over maximizing individual trade magnitude. He also utilizes fixed-time holding periods, systematically closing trades after a predefined count of three, five, or ten days to avoid overexposure to changing regimes.

Andrea Unger

The only individual to win the WCTC four times, Andrea Unger epitomizes the evolution from a struggling discretionary retail trader to a world-class systematic engineer. Unger's approach relies on extensive algorithmic automation and vast portfolio diversification. Early in his competitive career, his edge was built upon Opening Range Breakout (ORB) systems applied specifically to European indices like the DAX and American indices like the S&P 500. His logic involved measuring the high and low established during the initial hours of the trading session, recognizing that the resolution of this early volatility often dictates the directional trend for the remainder of the day. By placing stop orders outside this defined range, Unger captured intraday momentum while avoiding overnight holding risks.

Over time, Unger expanded his methodology to encompass a fully diversified portfolio containing trend-following, counter-trend, and seasonal bias systems distributed across highly volatile markets like crude oil and gold, as well as lower-volatility markets like the Euro FX. A hallmark of his approach is the "Plan B" contingency. Unger acknowledges that market regimes shift unpredictably; during the 2009 championship, his core trend-following models stagnated for six months. Because he had pre-planned alternative logic, he seamlessly pivoted to a suite of counter-trend algorithms, successfully capturing the necessary returns to win the title. His system development process is uniquely inverted; rather than building a theory-based setup, he tests a specific entry trigger against an extensive library of classical and volatility-based market patterns to empirically discover what precedes a successful outcome, thereby mapping the structural character of each individual asset.

Kevin Davey

A former aerospace engineer responsible for jet aircraft fuel pump quality control, Kevin Davey brought the rigorous standards of statistical process control to algorithmic trading, securing a WCTC victory in 2006. Davey's methodology, famously termed the "Strategy Factory," is defined by an extreme aversion to curve-fitting and historical overfitting. He insists that any algorithmic model must survive a gauntlet of out-of-sample testing, walk-forward analysis, and Monte Carlo simulations before it is deemed robust enough for live capital deployment.

Risk protection forms the core of his live execution architecture. Rather than relying solely on individual trade stop-losses, Davey engineers global portfolio fail-safes, including the "Daily Loss Limiter." If an algorithm's combined open and closed equity drops below a specific, pre-calculated monetary threshold during a single session, the system automatically flattens all positions and halts trading until the next day. This mechanical circuit breaker prevents catastrophic intraday spirals. Furthermore, he implements cooling-off periods within his algorithms, forcing the logic to wait for a predefined number of bars after a losing trade before re-entering the market, effectively neutralizing the algorithmic equivalent of "catching a falling knife".

Michael Cook

Michael Cook, who captured WCTC titles in 2007 and 2014, operates a fully systematic infrastructure driven partly by geographic necessity; residing in Australia made trading the highly liquid US market hours manually unfeasible. Cook's philosophy emphasizes that automation removes the trader's most significant liability: human conviction. He notes that the trades he subjectively perceives as terrifying or visually chaotic are historically his most profitable, whereas his highest-conviction discretionary setups frequently fail.

Cook's primary strategic insight lies in the sequencing of risk management. He argues that most retail traders calculate a fixed dollar amount they are willing to lose, and arbitrarily place a stop-loss order at that exact distance. Cook views this as a structural error. Instead, his algorithms identify where the stop-loss logically belongs based on market mechanics, support/resistance zones, and volatility expansion. Only after the market-dictated stop distance is calculated does the algorithm size the position to ensure the monetary exposure aligns with his risk parameters. Cook operates across 60 to 70 uncorrelated markets, leveraging the scale that only full automation can provide, and posits that the vast majority of untapped alpha resides in dynamic position sizing rather than entry optimization.

Stefano Serafini

Winning the 2017 WCTC with a 2172% return, Stefano Serafini focuses heavily on short-term intraday trading across commodities and indices. His decision to eliminate overnight holding periods is statistically driven; capturing smaller, intraday swings exponentially increases the sample size of trades, allowing his statistical edge to materialize more reliably while simultaneously neutralizing gap risk caused by overnight macroeconomic shocks.

Serafini employs a unique data-engineering technique to filter market noise: the 1440-minute bar. Because traditional daily bars in futures markets are often subject to artificial settlement price distortions, constructing a continuous 24-hour synthetic bar provides a much cleaner representation of the underlying macro trend. Operationally, Serafini manages a vast library of approximately 100 coded strategies but only deploys five to six live at any given time. His selection process ignores theoretical backtest correlations, relying entirely on the real-time, live-money equity curve correlation of the systems to determine which models are currently synchronized with the active market regime, ruthlessly rotating out algorithms that display performance decay.

Tim Rea

Tim Rea, a dominant force in the 2008 and 2011 E-mini divisions, executes a high-frequency, massive-scale approach, running over 100 automated strategies simultaneously across 25 different futures contracts. Rea is highly skeptical of low-frequency models, arguing that a system generating only 50 trades over a five-year period lacks the statistical significance necessary to distinguish genuine edge from random variance. Consequently, his portfolio is densely packed with short-term algorithms holding positions for one to three days, ensuring his backtests contain thousands of data points.

Rea's filtering process for deploying a strategy is exceedingly strict. A model must demonstrate a historical profit-to-drawdown ratio of approximately 10:1; the total absolute return must be ten times larger than the deepest historical peak-to-trough decline before

transaction costs are even considered. Furthermore, he scrutinizes the average profit per trade. Strategies that rely on microscopic inefficiencies are discarded, as Rea understands that real-world slippage and commissions will completely erode thin margins. ↴

Ivan Scherman

Ivan Scherman secured the 2023 WCTC title by generating a 491.9% net return while maintaining an incredibly controlled maximum drawdown of just -26.2% over a ten-month period. As a Chartered Market Technician (CMT), Scherman relies on purely algorithmic, data-mined patterns to extract probability edges. His flagship methodology centers on a specific mean-reversion setup derived from the relationship between the S&P 500 and the VIX volatility index. ↴

Scherman's algorithm triggers a long entry when the S&P 500 experiences a secondary market correction, making a lower low relative to a previous swing, but the VIX displays a structural divergence by failing to make a corresponding higher high. This lack of expanding volatility suggests that institutional hedging and put-buying have exhausted themselves. An additional requirement mandates that the S&P 500 must be overextended, closing more than two standard deviations below its 30-day moving average. The exit logic is entirely systematic: the trade is closed at a profit when the index reverts back above the 30-day moving average, or closed at a loss if the price continues to break down further beyond the Bollinger Bands. Through extensive backtesting dating back to 1957, Scherman verified that this specific mechanical setup boasts a win rate exceeding 75% and a profit factor of 3.16, providing an exceptionally stable foundation for aggressive compounding. ↴

David Trullas Vila

In stark contrast to the systematic automation favored by annual champions, David Trullas Vila dominated the quarterly WCTC sprint divisions in 2025 and 2026—achieving returns of 585%, 800%, and 2,222%—using intense, discretionary manual scalping. A former physiotherapist, Trullas Vila treats trading as a high-performance psychological endeavor, operating in brief, highly concentrated one-hour sessions. ↴

He completely rejects complex modern trading methodologies such as Order Flow analysis or Smart Money Concepts, relying instead on a stripped-down framework of four classical indicators: MACD, RSI, Volume, and Fibonacci. Trading almost exclusively the highly volatile NASDAQ futures on ultra-short timeframes (five-minute or five-second charts), he aligns all trades with the overarching macro trend, holding a strict long bias during bull markets to avoid the danger of counter-trend traps. His primary entry trigger is the formation of regular or hidden bullish divergences between price action and the MACD/RSI oscillators, occurring exactly at key Fibonacci retracement levels. To manage risk dynamically while scaling, he sizes his positions organically, adding one specific contract lot for every individual divergence signal detected on the chart. ↴

Chuck Hughes

A multi-time trading champion spanning decades, Chuck Hughes specializes in applying robust trend-following principles to the options and equities markets. Hughes developed the "Power Trend" system, which mandates that the underlying asset's one-month closing price must be cleanly above its 10-month simple moving average (SMA) to establish a bullish macro bias. Once the trend is confirmed, Hughes employs Keltner Channels to identify lower-risk entry points, explicitly waiting for the asset to experience a temporary pullback into the "Buy Zone" below the upper channel band. ↴

To translate this directional edge into outsized returns, Hughes leverages the options market using his proprietary "1% Rule." He strictly purchases deep in-the-money call options where the extrinsic time value represents less than 1% of the underlying stock price. By stripping away time value decay (theta), his options synthetically mirror the delta of the underlying stock, ensuring that if the stock price increases by merely 1%, the option trade reaches the breakeven threshold and begins compounding massive percentage returns through leverage. ↴

Psychological and Behavioral Patterns

The data across these disparate methodologies reveals a distinct psychological profile necessary to endure and conquer the WCTC. The cognitive architecture required for this level of performance often stems from formative experiences with severe market trauma. Michael Cook's deeply conservative sizing bias was forged during the chaotic destruction of the Asian financial crisis and the LTCM collapse in the late 1990s, permanently altering his perception of tail risk. Similarly, Kevin Davey's aerospace engineering background instilled a zero-tolerance policy for failure, translating the rigor of jet engine quality control directly into the out-of-sample stress testing of his code. ↴

A paramount psychological trait among champions is the complete detachment from outcome bias. As highlighted in performance psychology frameworks, amateur traders judge the quality of their decisions entirely by the financial result of the trade, leading to devastating emotional swings. Professional champions, however, score their performance based strictly on procedural compliance. If a trader follows their algorithms flawlessly and experiences a losing week, the process is graded as a success; the variance is mathematically expected. David Trullas Vila explicitly acknowledges the mental toll of aggressive leverage, utilizing neuroscience-based cognitive strategies to maintain emotional equilibrium during steep drawdowns, treating his trading desk with the same physiological respect as a professional athlete. Ultimately, the transition to full algorithmic automation by traders like Unger and Cook serves as a mechanical safeguard, protecting the portfolio from the trader's own fragile psychology during moments of extreme market distress. ↴

Risk Management and Position Sizing Frameworks

While entries and exits dictate the statistical edge, the mathematical engine that propels a portfolio to triple-digit returns is aggressive, dynamic position sizing. Trading championships require leveraging capital to the absolute threshold of mathematical efficiency without crossing into terminal ruin.

Optimal f (Ralph Vince)

Developed by mathematician Ralph Vince, Optimal f is a highly aggressive position-sizing algorithm designed to find the exact fraction of capital to risk on each trade to maximize the geometric growth rate—the Terminal Wealth Relative (TWR)—of a portfolio over time. The model calculates this fraction by assessing the historical sequence of trades and grounding the formula in the single largest historical loss experienced by the system. ↴

While mathematically brilliant in theory, it is exceptionally dangerous in practice. Andrea Unger warns that Optimal f represents a "mathematical cliff" rather than a practical target. If a system's Optimal f dictates risking 27% per trade, pushing the risk to 30% does not merely increase volatility; it destroys the geometric compounding effect, leading to rapid capital depletion. Furthermore, because the formula relies on a known historical maximum loss, the entire model is vulnerable to an unprecedented "black swan" event that produces a new, larger worst-case scenario. Consequently, champions utilize Optimal f to identify the absolute theoretical ceiling of their risk, typically choosing to trade at a heavily discounted fraction of that limit (e.g., Half-Kelly) to endure the inevitable psychological strain of the massive drawdowns the formula produces. ↴

Fixed Ratio Position Sizing (Ryan Jones)

Popularized by Ryan Jones, the Fixed Ratio method disconnects position sizing from total account equity and links it directly to the absolute dollar profit generated by the strategy. The core mechanism is the "Delta"—the predefined amount of profit required per contract to increase the position size by one additional unit. ↴

The formula is defined as:

$$N = 0.5 \times [(1 + 8 \times P/\Delta)^{0.5} + 1]$$

Where N represents the number of contracts, P is the accumulated net profit, and Δ is the Delta value. ↴

This methodology is highly favored in sprint competitions because it scales exposure aggressively (as the square root of profit) during winning streaks, allowing smaller accounts to compound rapidly without requiring massive initial capital. Conversely, when a drawdown occurs, the accumulated profit drops, and the algorithm swiftly decelerates the position size, providing a mechanical parachute. However, a significant limitation is that as the account grows exponentially, the fixed Delta becomes a bottleneck, forcing the trader to manually transition to fractional models to fully utilize the available capital base. ↴

Fixed Fractional Risk

The most sustainable and universally adopted framework among multi-year systematic champions, such as Michael Cook, is Fixed Fractional Risk. In this model, the trader commits a fixed percentage of total current equity (e.g., 2% or 5%) to the size of each trade.

Fractional sizing: In this model, the trader commits a rigid percentage of total current equity (e.g., 2% or 5%) to the risk of any single trade.

The calculation operates sequentially: First, the algorithm dictates the structural stop-loss distance based on market volatility (e.g., a multiple of the Average True Range). Second, the algorithm calculates the absolute dollar amount representing the fractional risk (e.g., 5% of a \$50,000 account = \$2,500). Finally, the dollar risk is divided by the stop-loss distance to determine the exact contract lot size. This structure ensures that capital is preserved during losing streaks, as the absolute dollar risk shrinks proportionately with the account balance, while organically expanding exposure as equity reaches new highs.

Performance Metrics and Drawdown Control

To maintain discipline and assess edge degradation, professional quantitative researchers rely on a suite of advanced diagnostic metrics rather than merely tracking cumulative profit and loss.

- Expectancy:** Expectancy calculates the average dollar amount or risk-multiple a trader can anticipate earning per execution over a large sample size. It is calculated as: $(\text{Win Rate} \times \text{Average Win}) - (\text{Loss Rate} \times \text{Average Loss})$. A strategy can possess a low win rate (e.g., 35%) and still boast a highly positive expectancy if the average win significantly dwarfs the average loss.
- Average R-Multiple:** Developed by Van Tharp, the R-Multiple standardizes performance by measuring each trade's outcome as a multiple of the initial risk taken. A trade risking \$500 that secures a \$1,500 profit is a +3R winner. Strategies that consistently produce an Average R-Multiple above +0.5R possess a deeply sustainable mathematical edge.
- Profit Factor:** The ratio of total gross profit to total gross loss. Institutional quants view a Profit Factor above 1.5 as robust, while metrics dropping below 1.0 indicate an immediate failure of edge.
- Drawdown Duration and Depth:** Traders like Kevin Davey monitor not only the depth of a peak-to-trough equity decline but its duration. A 15% drawdown recovered in a week represents normal variance; the exact same depth stretching over four months indicates a fundamental decoupling between the algorithm and the current market regime.

Common Patterns Across Successful Traders

Synthesizing the data across all analyzed WCTC champions reveals several absolute foundational principles required for outlier performance:

- Eradication of Human Intervention:** With the exception of ultra-short-term quarterly sprinters, almost all annual champions rely on rigid automation. Removing discretion protects the trader from emotional sabotage and enables the execution of complex strategies across dozens of markets simultaneously.
- Velocity of Trades (Short Holding Periods):** Annual champions overwhelmingly utilize intraday or short-term swing models holding assets for one to five days. High trade frequency rapidly expands the statistical sample size, allowing the mathematical edge to outpace random variance, while effectively neutralizing catastrophic overnight gap risks.
- Algorithmic Mercilessness:** Champions do not harbor emotional attachment to their code. If a live equity curve diverges significantly from the Monte Carlo simulation or breaches a predefined drawdown threshold, the system is permanently disabled without hesitation.
- Sizing as the True Alpha:** The prevailing consensus among veterans is that obsessing over perfect entry indicators yields diminishing returns. The true mechanism for alpha generation is deploying an imperfect entry paired with highly dynamic, volatility-adjusted position sizing.

Strategy Ideas for Python Conversion

Drawing directly from the verified methodologies of WCTC champions, the following conceptual architectures serve as the foundation for the Python algorithmic reconstruction:

- DAX Volatility Opening Range Breakout (Unger-Inspired):** Utilizing 15-minute intervals, calculate the absolute high and low established during the first 60 minutes of the primary trading session. Apply a volatility multiplier (e.g., 1.5x ATR) to the previous close to filter out false noise. Execute a breakout entry in the direction of the trend, applying a trailing stop and a hard time-based exit to flatten the portfolio prior to the daily close.
- S&P 500 Divergence Mean Reversion (Scherman-Inspired):** Construct a mean-reversion model identifying moments of acute panic. Trigger a long entry exclusively when the index price closes beyond two standard deviations below a 30-day moving average (Bollinger Band lower breach), provided a secondary volatility index (or ATR proxy) has failed to register a corresponding higher high.
- Williams %R Macro-Aligned Pullback:** Establish a macro-trend filter using the relationship between a 50-day and 200-day Simple Moving Average. During a confirmed bullish regime, wait for the 14-period Williams %R to plunge below the -80 threshold, indicating temporary oversold exhaustion. Execute a long entry upon the oscillator crossing back above -80, utilizing a structural stop-loss and a rapid "bailout" exit upon reaching the -20 overbought threshold.
- Trend Continuation with Aggressive Pyramiding:** Implement a 20-period Donchian Channel to capture structural breakouts. Size the initial position conservatively using an ATR-derived fractional risk model. If the asset trends favorably by a distance of 1R (one unit of initial risk), trigger a pyramiding function to aggressively add a secondary lot, simultaneously trailing the global stop-loss to the breakeven threshold to lock in capital preservation.

Verified Data versus Inferred Logic

To maintain academic rigor, it is essential to delineate between empirical facts verified by public audit and proprietary parameters inferred through structural analysis.

Data Category	Verified Empirical Facts	Inferred / Reconstructed Logic
Performance Data	Final net percentages, official rankings, and WCTC dates.	Real-time equity curve trajectories during the middle of the contest.
Strategy Archetypes	Unger's DAX breakouts, Williams' COT usage, Scherman's VIX divergence, Hughes' Keltner trades.	The precise lookback periods, multipliers, and moving average lengths utilized in live execution.
Risk Frameworks	Reliance on Optimal f , Fixed Ratio, and Daily Loss Limiters.	The exact decimal fraction of Kelly or the specific monetary Delta implemented by the champions.
Trade Execution	Usage of TradeSync AutoTrade technology for follower replication.	Proprietary slippage models and latency optimization techniques used on their private servers.

Limitations of the Research

A critical limitation inherent in studying WCTC outcomes is the pervasive influence of survivorship bias. Industry analysts point out that the WCTC is a private, unregulated competition that permits participants to fund and operate an unlimited number of accounts simultaneously. An aggressive competitor could theoretically fund ten separate minimum-balance accounts, apply hyper-leveraged Optimal f sizing to each using slightly divergent parameters, incinerate nine accounts to zero, and secure the championship title with the single surviving account that captured a tail-event variance.

Because the WCTC leaderboards exclusively display the highest-performing accounts and shield burned accounts from public scrutiny, the triple-digit returns do not represent sustainable institutional risk-adjusted benchmarks. Rather, they highlight the statistical upper limits of aggressive leverage applied over brief timeframes. While the structural entry, exit, and filtering logic of these champions is fundamentally sound, replicating their championship leverage profiles with long-term retirement or institutional capital would almost certainly result in catastrophic ruin.

Quantitative Strategy Templates (Python Logic)

The following Python implementation translates the synthesized research into a production-ready, object-oriented backtesting architecture.

The `BacktestEngine` class is designed to process standard OHLCV data. It calculates all necessary technical overlays, including Moving Averages, Donchian Channels, Average True Range (ATR), and the Williams %R momentum oscillator. The engine is fully modular, allowing the user to select between distinct strategies: trend-following moving average crossovers, Donchian breakouts, Williams %R mean reversion, and a volatility breakout proxy for Opening Range strategies.

Crucially, the code enforces institutional-grade risk management. It utilizes ATR-based stop-losses derived from prevailing market structure rather than arbitrary percentages. The `simulate_trading` function executes Fixed Fractional position sizing, calculates transaction costs and slippage, and completely eliminates look-ahead bias by shifting signal execution to the subsequent open. Furthermore, a pyramiding flag is included to model the aggressive scale-in techniques favored by contest winners, alongside a `walk_forward_optimization` function to mimic Kevin Davey's rigorous out-of-sample stress testing.

Python Implementation

Python

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from itertools import product
import warnings
warnings.filterwarnings('ignore')

class BacktestEngine:
    def __init__(self, data, initial_capital=10000, risk_per_trade=0.02, commission=1.0, slippage=1.0):
        """
        Initializes the robust backtesting engine.
        data: pandas DataFrame with DatetimeIndex and 'Open', 'High', 'Low', 'Close'
        """
        self.data = data.copy()
        self.initial_capital = initial_capital
        self.risk_per_trade = risk_per_trade
        self.commission = commission # Flat fee per executed trade
        self.slippage = slippage # Slippage penalty in price units

    def _calculate_atr(self, period=14):
        """Calculates the Average True Range (ATR) for volatility measurement."""
        high_low = self.data['High'] - self.data['Low']
        high_close = np.abs(self.data['High'] - self.data['Close'].shift())
        low_close = np.abs(self.data['Low'] - self.data['Close'].shift())
        ranges = pd.concat([high_low, high_close, low_close], axis=1)
        true_range = np.max(ranges, axis=1)
        return true_range.rolling(period).mean()

    def apply_indicators(self, params):
        """Generates all required technical indicators based on dynamic parameters."""
        # Moving Averages for macro trend filtering
        self.data['SMA_Fast'] = self.data['Close'].rolling(window=params.get('sma_fast', 10)).mean()
        self.data['SMA_Slow'] = self.data['Close'].rolling(window=params.get('sma_slow', 50)).mean()

        # Donchian Channels for structural breakouts
        dc_period = params.get('donchian_period', 20)
        self.data['Donchian_High'] = self.data['High'].rolling(window=dc_period).max().shift(1)
        self.data['Donchian_Low'] = self.data['Low'].rolling(window=dc_period).min().shift(1)

        # Williams %R Oscillator (Scale: 0 to -100)
        wr_period = params.get('wr_period', 14)
        highest_high = self.data['High'].rolling(window=wr_period).max()
        lowest_low = self.data['Low'].rolling(window=wr_period).min()
        self.data['Williams_R'] = ((highest_high - self.data['Close']) / (highest_high - lowest_low)) * -100

        # Volatility & ATR mapping
        self.data['ATR'] = self._calculate_atr(params.get('atr_period', 14))

    def generate_signals(self, strategy_type='donchian'):
        """Computes trade signals. Strict shifting is applied to prevent look-ahead bias."""
        self.data['Signal'] = 0

        if strategy_type == 'donchian':
            # Classic trend breakout logic
            long_cond = self.data['Close'] > self.data['Donchian_High']
            short_cond = self.data['Close'] < self.data['Donchian_Low']

        elif strategy_type == 'moving_average':
            # Moving average cross logic
            long_cond = self.data['SMA_Fast'] > self.data['SMA_Slow']
            short_cond = self.data['SMA_Fast'] < self.data['SMA_Slow']

        elif strategy_type == 'williams_r':
            # Mean reversion strictly aligned with the macro trend
            macro_bull = self.data['Close'] > self.data['SMA_Slow']
            macro_bear = self.data['Close'] < self.data['SMA_Slow']

            # Long trigger: %R crosses back above -80. Short trigger: crosses below -80.
            long_cond = macro_bull & (self.data['Williams_R'].shift(1) < -80) & (self.data['Williams_R'] >= -80)
            short_cond = macro_bear & (self.data['Williams_R'].shift(1) > -20) & (self.data['Williams_R'] <= -20)

        elif strategy_type == 'volatility_breakout':
            # Dynamic volatility channel proxy for Opening Range Breakout
            k = 1.5
            upper_band = self.data['Close'].shift(1) + (k * self.data['ATR'].shift(1))
            lower_band = self.data['Close'].shift(1) - (k * self.data['ATR'].shift(1))
            long_cond = self.data['Close'] > upper_band
            short_cond = self.data['Close'] < lower_band

        # Map signals: 1 for Long, -1 for Short
        self.data.loc[long_cond, 'Signal'] = 1
        self.data.loc[short_cond, 'Signal'] = -1

        # Shift signal to execute on the *following* bar's Open
        self.data['Execution_Signal'] = self.data['Signal'].shift(1)

    def simulate_trading(self, use_pyramiding=False):
        """Simulates historical execution, modeling fractional sizing, slippage, and stop-losses."""
        capital = self.initial_capital
        equity_curve = [capital]
        trade_log = []

        position = 0 # 1 (Long), -1 (Short)
        entry_price = 0
        entry_atr = 0
        units = 0

        for i in range(1, len(self.data)):
            row = self.data.iloc[i]
            prev_row = self.data.iloc[i-1]
            signal = row['Execution_Signal']
```

```

# 1. Process Stop Loss / Exits First
if position != 0:
    stop_dist = entry_atr * 2 # Standard 2 ATR Stop distance
    stop_price = entry_price - stop_dist if position == 1 else entry_price + stop_dist

    # Verify if extreme prices breached the stop level
    if (position == 1 and row['Low'] <= stop_price) or (position == -1 and row['High'] >= stop_price):
        exit_price = stop_price - self.slippage if position == 1 else stop_price + self.slippage
        pnl = (exit_price - entry_price) * units * position - self.commission
        capital += pnl
        trade_log.append({'Type': 'Loss', 'Pnl': pnl, 'Risk': stop_dist * units})
        position, units = 0, 0
        equity_curve.append(capital)
        continue

# 2. Pyramiding Logic (Aggressively adding to winning positions)
if position != 0 and use_pyramiding:
    favorable_move = row['Close'] - entry_price if position == 1 else entry_price - row['Close']
    if favorable_move > entry_atr: # Asset has moved 1 ATR in favor
        add_units = units * 0.5 # Scale in with 50% of original sizing
        units += add_units
        entry_price = (entry_price + row['Close']) / 2 # Adjust average basis
        capital -= self.commission # Deduct transaction cost for new tier

# 3. Process New Entries and Reversals
if signal == 1 and position != 1:
    if position == -1: # Liquidate existing short
        exit_price = row['Open'] + self.slippage
        pnl = (entry_price - exit_price) * units - self.commission
        capital += pnl
        trade_log.append({'Type': 'Win' if pnl > 0 else 'Loss', 'Pnl': pnl, 'Risk': entry_atr * 2 * units})

    # Execute Long Position
    position = 1
    entry_price = row['Open'] + self.slippage
    entry_atr = prev_row['ATR'] if pd.notna(prev_row['ATR']) else entry_price * 0.01

    # Calculate Fixed Fractional Sizing based on structural risk
    risk_amount = capital * self.risk_per_trade
    risk_per_unit = entry_atr * 2
    units = max(1, risk_amount / risk_per_unit) if risk_per_unit > 0 else 1
    capital -= self.commission

elif signal == -1 and position != -1:
    if position == 1: # Liquidate existing long
        exit_price = row['Open'] - self.slippage
        pnl = (exit_price - entry_price) * units - self.commission
        capital += pnl
        trade_log.append({'Type': 'Win' if pnl > 0 else 'Loss', 'Pnl': pnl, 'Risk': entry_atr * 2 * units})

    # Execute Short Position
    position = -1
    entry_price = row['Open'] - self.slippage
    entry_atr = prev_row['ATR'] if pd.notna(prev_row['ATR']) else entry_price * 0.01

    risk_amount = capital * self.risk_per_trade
    risk_per_unit = entry_atr * 2
    units = max(1, risk_amount / risk_per_unit) if risk_per_unit > 0 else 1
    capital -= self.commission

# 4. Mark portfolio to market
if position == 1:
    mtm = capital + ((row['Close'] - entry_price) * units)
elif position == -1:
    mtm = capital + ((entry_price - row['Close']) * units)
else:
    mtm = capital

equity_curve.append(mtm)

self.data['Equity'] = equity_curve
self.trade_log = pd.DataFrame(trade_log)
return self.calculate_metrics()

def calculate_metrics(self):
    """Computes advanced quantitative performance and risk-adjusted metrics."""
    equity = self.data['Equity']
    if len(self.trade_log) == 0:
        return {'CAGR': 0, 'Sharpe': 0, 'Max DD': 0, 'Win Rate': 0, 'Profit Factor': 0, 'Expectancy': 0}

    # Absolute temporal calculations
    days = (self.data.index[-1] - self.data.index[0]).days
    years = days / 365.25

    # CAGR and Drawdown
    total_return = equity.iloc[-1] / equity.iloc[0]
    cagr = (total_return ** (1 / years)) - 1 if years > 0 else 0
    rolling_max = equity.cummax()
    drawdown = (equity - rolling_max) / rolling_max
    max_dd = drawdown.min()

    # Risk-Adjusted Ratios
    daily_returns = equity.pct_change().dropna()
    risk_free_rate = 0.02 / 252
    sharpe = np.sqrt(252) * (daily_returns.mean() - risk_free_rate) / daily_returns.std() if daily_returns.std() != 0 else 0
    downside_std = daily_returns[daily_returns < 0].std()
    sortino = np.sqrt(252) * (daily_returns.mean() - risk_free_rate) / downside_std if downside_std != 0 else 0

    # Granular Trade Metrics
    wins = self.trade_log[self.trade_log['Pnl'] > 0]
    losses = self.trade_log[self.trade_log['Pnl'] <= 0]
    win_rate = len(wins) / len(self.trade_log)
    gross_profit = wins['Pnl'].sum()
    gross_loss = abs(losses['Pnl'].sum())
    profit_factor = gross_profit / gross_loss if gross_loss != 0 else np.inf

    avg_win = wins['Pnl'].mean() if len(wins) > 0 else 0
    avg_loss = abs(losses['Pnl'].mean()) if len(losses) > 0 else 0
    expectancy = (win_rate * avg_win) - ((1 - win_rate) * avg_loss)

    # System R-Multiple Validation
    self.trade_log['R_Multiple'] = self.trade_log['Pnl'] / self.trade_log['Risk']
    avg_r_multiple = self.trade_log['R_Multiple'].mean()

    return {
        'Total Trades': len(self.trade_log),
        'CAGR (%)': round(cagr * 100, 2),
        'Max Drawdown (%)': round(max_dd * 100, 2),
        'Sharpe Ratio': round(sharpe, 2),
        'Sortino Ratio': round(sortino, 2),
    }

```

```

        'Win Rate (%)': round(win_rate * 100, 2),
        'Profit Factor': round(profit_factor, 2),
        'Expectancy ($)': round(expectancy, 2),
        'Avg R-Multiple': round(avg_r_multiple, 2)
    }

def plot_equity_curve(self):
    """Generates a visualization of the portfolio equity curve and drawdown profile."""
    fig, ax1 = plt.subplots(figsize=(12, 6))
    ax1.plot(self.data.index, self.data['Equity'], color='blue', label='Portfolio Equity')
    ax1.set_ylabel('Capital ($)', color='blue')
    ax1.set_title('Strategy Equity Curve & Drawdown Profile')
    ax1.grid(True, alpha=0.3)
    plt.legend(loc='upper left')
    return fig

def walk_forward_optimization(data, param_grid, train_size=252, test_size=63):
    """
    Simulates Kevin Davey's 'Strategy Factory' out-of-sample stress testing.
    train_size: approximately 1 trading year.
    test_size: approximately 1 trading quarter.
    """
    results = []
    keys = param_grid.keys()
    combinations = [dict(zip(keys, v)) for v in product(*param_grid.values())]

    start_idx = 0
    while start_idx + train_size + test_size < len(data):
        train_data = data.iloc[start_idx : start_idx + train_size]
        test_data = data.iloc[start_idx + train_size : start_idx + train_size + test_size]

        best_param = None
        best_sharpe = -np.inf

        # In-Sample Optimization Phase
        for params in combinations:
            engine = BacktestEngine(train_data)
            engine.apply_indicators(params)
            engine.generate_signals('donchian')
            metrics = engine.simulate_trading()

            if metrics['Sharpe Ratio'] > best_sharpe:
                best_sharpe = metrics['Sharpe Ratio']
                best_param = params

        # Out-of-Sample Validation Phase
        test_engine = BacktestEngine(test_data)
        test_engine.apply_indicators(best_param)
        test_engine.generate_signals('donchian')
        test_metrics = test_engine.simulate_trading()

        results.append({
            'Period': f'{test_data.index[0].date()} to {test_data.index[-1].date()}',
            'Optimal Parameters': best_param,
            'OOS Sharpe': test_metrics['Sharpe Ratio'],
            'OOS Return (%)': test_metrics['CAGR (%)']
        })

        # Advance the rolling window
        start_idx += test_size

    return pd.DataFrame(results)

```

Next Steps for Improving the Strategies

To elevate the provided Python architecture from an analytical backtesting tool into a production-grade algorithmic deployment suite capable of WCTC-level execution, the following enhancements are required:

- Regime Detection and Dynamic Allocation:** Following the methodologies of Andrea Unger and Stefano Serafini, the system must integrate an unsupervised regime-detection filter—such as a Hidden Markov Model or ADX thresholding. This layer would automatically shift capital allocation between the `donchian` (trend-following) logic and `williams_r` (mean-reversion) logic in real-time, depending on prevailing market volatility structures.
- Tick-Level Ingestion and Synthetic Bars:** The current engine processes standard OHLC data structures. To fully replicate Serafini's specific data edge, functionality must be constructed to ingest raw tick-level data and programmatically resample it into custom 1440-minute rolling bars, neutralizing the artificial distortions caused by exchange settlement periods.
- Monte Carlo Ruin Simulations:** The `BacktestEngine` should be expanded to include a Monte Carlo module that randomly resamples the generated `trade_log` 10,000 times with replacement. By plotting the distribution of terminal equity across these permutations, developers can mathematically quantify the true risk of ruin at various Fixed Fractional sizing limits, satisfying Kevin Davey's requirement for absolute statistical robustness.
- Integration of 'Delta' Fixed Ratio Sizing:** The position sizing module within the `simulate_trading` function must be refactored to allow programmatic toggling between standard Fixed Fractional allocation and Ryan Jones' Fixed Ratio calculation. This will empower the researcher to empirically test whether scaling linearly or exponentially (as a function of the square root of profit) yields superior geometric growth for specific target assets.

Sources used in the report ^

-  justmarkets.com World Cup Trading Championships Overview and Insights - JustMarkets
-  kjttradingsystems.com Award-Winning Algorithmic Trading Course & Strategies | Kevin Davey
-  ungeracademy.com The Ultimate Guide to Joining the World Cup Trading Championships: Tips from a 4-Time Winner - Unger Academy EN
-  worldcupchampionships.com World Cup Trading Championship Standings
-  insider-week.com World Cup Trading Championships – a great success not once but twice
-  worldcupchampionships.com 2024 quarterly finals - World Cup Trading Championships
-  worldcupchampionships.com World Cup Trading Championship Historical Standings
-  moneyshow.com Chuck Hughes - MoneyShow
-  worldcupchampionships.com Become a Professional Trader - World Cup Trading Championships
-  ib.stonex.com World Cup Advisor - Futures Systems Trading - StoneX Futures
-  worldcupadvisor.com About Us - World Cup Advisor
-  pictureperfectportfolios.com How To Invest Like Larry Williams: When Systematic Mechanics Beat Market Gut Feelings
-  market-bulls.com Larry Williams Trading Strategy Insights | MarketBulls
-  emini-watch.com Larry Williams Trading Strategy: Lessons from the Master - Emini-Watch
-  scribd.com Insights from Larry Williams on Trading | PDF | Foreign Exchange Market | Stocks - Scribd

[fidelity.com](#) Williams %R - Fidelity Investments

[en.wikipedia.org](#) Williams %R - Wikipedia

[altrady.com](#) Larry Williams Trading Strategy | Notable Techniques - Altrady

[ireallytrade.com](#) Larry Williams Hall of Fame | I Really Trade

[ungeracademy.com](#) How to Win the World Cup Trading Championships: 5 Practical Tips from the Only 4-Time Champion - Unger Academ...

[ungeracademy.com](#) My Journey To Winning The Trading Championships - Unger Academy EN

[bettersystemtrader.com](#) Andrea Unger on Building Trading Strategies That Work - World Cup Champion

[benzinga.com](#) DAX Trading: Exploring an Opening Range Breakout Strategy - Code and Optimizations

[ungeracademy.com](#) DAX First Hour Strategy (Last Year Performance) - Unger Academy EN

[bettersystemtrader.com](#) Trade Entry Techniques for Systematic Traders

[bettersystemtrader.com](#) How to Detect a Failing Trading Strategy | Better System Trader

[twoblokestrading.com](#) Algo Trading Masterclass: Kevin Davey's Secrets To Consistency Ep.7 Season 3

[ultimate-algo-trading.teachable.com](#) Strategy Factory Workshop | Ultimate Algo Trading

[kijtradingsystems.medium.com](#) Algorithmic Trading Tip — Building Risk Protection Into Your Trading | by Kevin Davey

[bettersystemtrader.com](#) Position Sizing and Stops for Algo Traders

[youtube.com](#) 112: Intraday Daytrading with Stefano Serafini [AUDIO ONLY] - YouTube

[ungeracademy.com](#) Trading Hero: Stefano Serafini - World Trading Champion 2017 - Unger Academy EN

[bettersystemtrader.com](#) Trading 100+ Algo Strategies: Tim Rea on Managing Multiple Systems

[worldcupadvisor.com](#) Follow Ivan Scherman - World Cup Advisor

[forbes.com.au](#) Ivan Scherman: Meet the man behind the money machine - Forbes Australia

[vikingen.se](#) The trader with 75 percent profits - Börsprogram för dig som vill bli ännu rikare - Vikingen.se

[worldcupchampionships.com](#) World Cup Trading Championship Standings - Version 2

[youtube.com](#) MACD, RSI, Volume and Fibonacci: The World Champion's System | David Trullas

[youtube.com](#) "Billionaire trading is a lie: I work one hour at maximum intensity." - David Trullas - YouTube

[chuckhughesonline.com](#) Chuck Hughes - Looking for a wealth-building opportunity?

[barchart.com](#) Unlock the Power of High-Accuracy Option Trading - Barchart.com

[chuckhughesonline.com](#) Power Up Your Portfolio with NextEra? - Chuck Hughes

[journalplus.co](#) Weekly and Monthly Trade Reviews: Complete Framework - JournalPlus

[journalplus.co](#) Building a Trading Performance Dashboard - JournalPlus

[youtube.com](#) 4-Time World Trading Champion Reveals His Method | David Trullas and Pietro Valastro

[turtlettrader.com](#) Optimal F by Ralph Vince: Money Management Theory vs. Trend Following Reality

[prorealcode.com](#) Ralph Vince's Optimal F Positioning Sizing - ProRealTime Forum - ProRealCode

[bettersystemtrader.com](#) Ralph Vince on Position Sizing, Optimal f, Diversification and Risk Management - Better System Trader

[quantpedia.com](#) Beware of Excessive Leverage - Introduction to Kelly and Optimal F - QuantPedia

[adaptrade.com](#) Fixed Ratio Position Sizing : Stock Trading Online : Day Trader Software

[daytrading.com](#) Fixed Ratio Betting in Trading - DayTrading.com

[www.2wealth-lab.com](#) Fixed Ratio - Wealth-Lab Wiki

[mymaamarkets.com](#) Mastering Risk: 5 Position Sizing Strategies Every Trader Should Know - My MAA Markets

[tickerly.net](#) How to Evaluate Trading Results Like a Pro - Tickerly

[tradezella.com](#) How to Analyze Your Trading Performance (The Data-Driven Approach) - TradeZella

[macro-ops.com](#) Mark D. Cook's Trading Strategy Explained | Macro Ops

[reddit.com](#) How to 'win' the Trading World Cup with zero trading skill : r/Daytrading - Reddit

[reddit.com](#) Does Robbin's World Cup have legitimacy? : r/Daytrading - Reddit

Sources read but not used in the report ▾

Thoughts ▾